

Lecture 21-23

Chapter: 05

Conditional Distributions, Conditional Expectations and Independence

Math 4441: Probability and Statistics

Reference: Goodman & Yates – Introduction to Probability and Stochastic Process, 3rd Edition

Introduction

- A conditional distribution is a complete probability model, and it follows the axioms of probability
- It is called conditional distribution because it is conditioned on the availability of some partial information
- One of the most important topics in probability, because
 - We are often interested in probabilities or expectations with some partial information available
 - It is often useful to assume the availability of some partial information
- Conditional distributions extend the conditional probability explained in Chapter 1 for Random variables

Conditional probabilities

Conditional probability

$$P[A|B] = \frac{P[AB]}{P[B]}, P[B] > 0$$

Product Rule or Law of Multiplication

$$P[AB] = P[A|B]P[B] = P[B|A]P[A]$$

Sum Rule or Law of Total Probability

$$P[A] = \sum_{i=1}^n P[A|B_i]P[B_i]$$

Bayes Theorem

$$P[B_i|A] = \frac{P[A|B_i]P[B_i]}{\sum_{i=1}^n P[A|B_i]P[B_i]}$$

Independence

$$P[AB] = P[A]P[B]$$

Conditional Distribution

1. Conditioned on an event

Experiment: Roll a fair six-sided die

$$S_X = \{1, \dots, 6\}$$

$A \triangleq \{\text{event that the outcome is an even number}\}$

$$P[A] = \frac{1}{2}$$

$$P_{X|A}(x) = \frac{P_X(x)}{P[A]}, P[A] > 0$$

x	1	2	3	4	5	6
$P_X(x)$						

$$A = \{2, 4, 6\}$$

$x \cap A$	2	4	6
$P_{X A}(x)$			

$$P_X(x) = \frac{1}{6}, \quad x = 1, \dots, 6$$

$$P_{X|A}(x) = \begin{cases} \frac{1}{3}, & x = 2, 4, 6 \\ 0, & \text{otherwise} \end{cases}$$

2. Conditioned on another Random Variables

$$P_{X|Y}(x|y) = \frac{P_{XY}(x,y)}{P_Y(y)}$$

$$P_{Y|X}(y|x) = \frac{P_{XY}(x,y)}{P_X(x)}$$

$P_{XY}(x, y)$	$y = 1$	$y = 2$	$P_X(x)$
$x = 1$			
$x = 2$			
$P_Y(y)$			

$$P_{X|Y}(x|y) = \frac{P_{XY}(x, y)}{P_Y(y)}$$

$P_{X Y}(x y)$	$y = 1$	$y = 2$
$x = 1$		
$x = 2$		

$P_{Y X}(y x)$	$y = 1$	$y = 2$
$x = 1$		
$x = 2$		

Product Rule or Law of Multiplication

$$P_{XY}(x, y) = P_{X|Y}(x|y)P_Y(y)$$

$$P_{XY}(x, y) = P_{Y|X}(y|x)P_X(x)$$

Example 6.11. Professor May B. Right often has her facts wrong, and answers each of her students questions incorrectly with probability $\frac{1}{4}$, independently of other questions. In each lecture, May is asked 0, 1, and 2 questions with equal probability of $\frac{1}{3}$. Let X and Y represent the number of questions May is asked in a lecture, and the number of questions she answers wrong in that lecture, respectively.

1. Find the joint PMF of X and Y , $P_{XY}(x, y)$.
2. Find the probability of at least one wrong answer.

$X \sim$

$$P_X(x) =$$

$Y|X = x \sim$

$$P_{Y|X}(y|x) =$$

$$P_{XY}(x, y) = P_{Y|X}(y|x)P_Y(y)$$

$P_{Y X}(y x)$	$y = 0$	$y = 1$	$y = 2$	$P_X(x)$
$x = 0$				
$x = 1$				
$x = 2$				

$P_{XY}(x, y)$	$y = 0$	$y = 1$	$y = 2$	$P_X(x)$
$x = 0$				
$x = 1$				
$x = 2$				
$P_Y(y)$				

$P_{XY}(x, y)$	$y = 0$	$y = 1$	$y = 2$	$P_X(x)$
$x = 0$	$\frac{16}{48}$	0	0	$\frac{16}{48}$
$x = 1$	$\frac{12}{48}$	$\frac{4}{48}$	0	$\frac{16}{48}$
$x = 2$	$\frac{9}{48}$	$\frac{6}{48}$	$\frac{1}{48}$	$\frac{16}{48}$
$P_Y(y)$	$\frac{37}{48}$	$\frac{10}{48}$	$\frac{1}{48}$	1.0

$$\begin{aligned}
 P[A] &= P_{XY}(1, 1) + P_{XY}(1, 2) + P_{XY}(2, 2) \\
 &= \frac{4}{48} + \frac{6}{48} + \frac{1}{48} \\
 &= \frac{11}{48}.
 \end{aligned}$$

Example 6.12. Consider four independent rolls of a 6-sided die. Let X be the number of 1's, and Y be the number of 2's obtained. What is the joint PMF of X and Y ?

Solution. We use the multiplication rule to calculate the joint PMF $P_{XY}(x, y)$, which is the product of the following quantities.

1. The conditional PMF of X , $P_{X|Y}(x|y)$, conditioned on the value of Y , $Y = y$.
2. The marginal PMF of Y , $P_Y(y)$.

$X \sim$

$$P_X(x) =$$

$Y|X = x \sim$

$$P_{X|Y}(x|y) =$$

$$P_{XY}(x, y) = P_{X|Y}(x|y)P_Y(y) = \binom{4-y}{x} \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{4-y-x} \binom{4}{y} \left(\frac{1}{6}\right)^y \left(\frac{5}{6}\right)^{4-y}$$

Example 6.13. If X_1 and X_2 are two independent binomial random variables with respective parameters (n_1, p) and (n_2, p) , calculate the conditional PMF of X_1 , given that $X_1 + X_2 = m$.

$$\begin{aligned} P[X_1 = k | X_1 + X_2 = m] &= \frac{P[X_1 = k, X_1 + X_2 = m]}{P[X_1 + X_2 = m]} \\ &= \frac{P[X_1 = k, X_2 = m - X_1]}{P[X_1 + X_2 = m]} \\ &= \frac{P[X_1 = k, X_2 = m - k]}{P[X_1 + X_2 = m]} \end{aligned}$$

Since X_1 has the Binomial distribution with parameters (n_1, p) , therefore,

$$P[X_1 = k] = \binom{n_1}{k} p^k q^{n_1-k}$$

Similarly, for X_2 , we have

$$P[X_2 = m - k] = \binom{n_2}{m - k} p^{m-k} q^{n_2-m+k}$$

$$P[X_1 + X_2 = m] = \binom{n_1 + n_2}{m} p^m q^{n_1+n_2-m}.$$

$$\begin{aligned}
P[X_1 = k | X_1 + X_2 = m] &= \frac{\binom{n_1}{k} p^k q^{n_1-k} \binom{n_2}{m-k} p^{m-k} q^{n_2-m+k}}{\binom{n_1+n_2}{m} p^m q^{n_1+n_2-m}} \\
&= \frac{\binom{n_1}{k} \binom{n_2}{m-k}}{\binom{n_1+n_2}{m}}.
\end{aligned}$$

Sum Rule or Law of Total Probability

$$\begin{aligned} P_X(x) &= \sum_y P_{XY}(x, y) \\ &= \sum_y P_{X|Y}(x|y) P_Y(y) \end{aligned}$$

$P_{X Y}(x y)$	$y = 1$	$y = 2$	$P_X(x)$
$x = 1$	$\frac{4}{6}$	$\frac{3}{4}$	
$x = 2$	$\frac{2}{6}$	$\frac{1}{4}$	
$P_Y(y)$	$\frac{6}{10}$	$\frac{4}{10}$	

$$P_X(x) = \sum_y P_{XY}(x|y)P_Y(y)$$

$$P_X(1) = P_{X|Y}(1|1)P_Y(1) + P_{X|Y}(1|2)P_Y(2)$$

$$P_X(2) = P_{X|Y}(2|1)P_Y(1) + P_{X|Y}(2|2)P_Y(2)$$

Conditional Expectation

$$E[X] = \sum_x x P_X(x)$$

$$E[X|A] = \sum_x x P_{X|A}(x).$$

$P_{X Y}(x y)$	$y = 1$	$y = 2$	$P_X(x)$
$x = 1$	$\frac{4}{6}$	$\frac{3}{4}$	$\frac{7}{10}$
$x = 2$	$\frac{2}{6}$	$\frac{1}{4}$	$\frac{3}{10}$

$$E[X|Y = 1] = 1 \cdot \frac{4}{6} + 2 \cdot \frac{2}{6} = \frac{8}{6}$$

$$E[X|Y = 2] = 1 \cdot \frac{3}{4} + 2 \cdot \frac{1}{4} = \frac{5}{4}$$

Law of Total Expectation

$$E[X] = \sum_y E[X|Y = y]P_Y(y)$$

$$E[Y] = \sum_x E[Y|X = x]P_X(x)$$

Example 6.17. Sam will read either one chapter of his probability book or one chapter of his history book. If the number misprints in a chapter of the probability book is Poisson distributed with mean 2 and if the number of misprints in the history chapter is Poisson distributed with mean 5. Then assuming Sam is equally likely to choose either book, what is the expected number of misprints he will come across?

Solution. Let X denote number of misprints, and

$$Y = \begin{cases} 1, & \text{if Sam chooses his history book} \\ 2, & \text{if Sam chooses his probability book} \end{cases}$$

Then,

$$\begin{aligned} E[X] &= E[X|Y = 1]P[Y = 1] + E[X|Y = 2]P[Y = 2] \\ &= 5 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} \\ &= \frac{7}{2}. \end{aligned}$$

Example 6.19. A miner is trapped in a mine containing three doors. The first door leads to a tunnel that takes him to safety after two hours of travel. The second door leads to a tunnel that returns him to the mine after three hours of travel. The third door leads to a tunnel that leads him to the mine after five hours. Assuming that the miner is at all times equally likely to choose any one of these doors, what is the expected length of time until the miner reaches safety?

$$\begin{aligned} E[X] &= \frac{1}{3}(E[X|Y = 1] + E[X|Y = 2] + E[X|Y = 3]) \\ &= \frac{1}{3}(2 + 3 + E[X] + 5 + E[X]) \end{aligned}$$

$$E[X] - \frac{2}{3}E[X] = \frac{10}{3}$$

$$E[X] = 10.$$

Example 6.18 (The mean and variance of Geometric Distribution). A coin having probability p of coming up head is to be successfully flipped until the first head appears. What is the expected number of flips required and the variance of number of flips?

$$P_X(x) = (1 - p)^{k-1}p \quad k \geq 1$$

$$E[X] = \sum_{k=1}^{\infty} k(1 - p)^{k-1}p, \text{ and} \quad Var[X] = \sum_{k=1}^{\infty} (k - E[X])^2(1 - p)^{k-1}p$$

$$Y = \begin{cases} 1, & \text{if the first flip results in a head} \\ 0, & \text{if the first flip results in a tail} \end{cases}$$

$$\begin{aligned} E[X] &= E[X|Y = 1]P[Y = 1] + E[X|Y = 0]P[Y = 0] \\ &= pE[X|Y = 1] + (1 - p)E[X|Y = 0] \end{aligned}$$

$$E[X|Y = 1] = 1$$

$$E[X|Y = 0] = E[1 + X] = 1 + E[X]$$

$$\begin{aligned} E[X] &= p \cdot 1 + (1 - p) \cdot (1 + E[X]) \\ &= p + 1 - p + E[X] - p \cdot E[X] \end{aligned}$$

$$\text{or, } p \cdot E[X] = 1$$

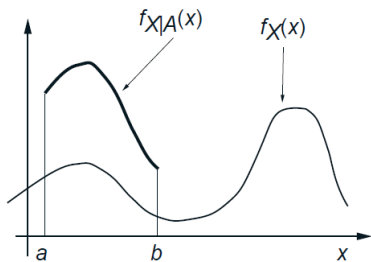
$$\text{or, } E[X] = \frac{1}{p}.$$

Continuous Conditional Distribution and Expectations

Conditioned on an Event

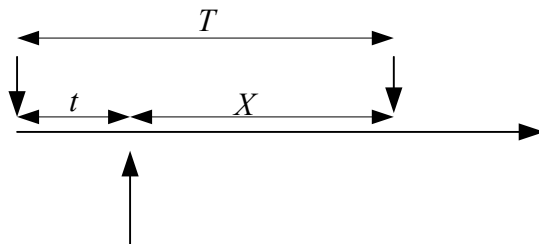
Definition 6.11. For a continuous random variable X with PDF $f_X(x)$ and an event $A \in S_X$ with $P[A] > 0$, where S_X represents the range of values of X , the conditional PDF of X conditioned on A is given by

$$f_{X|A}(x) = \begin{cases} \frac{f_X(x)}{P[A]}, & x \in A; \\ 0, & \text{otherwise.} \end{cases}$$



$$\begin{aligned} F_{X|A}(x) &= P[X \leq x|A] \\ &= \int_{-\infty}^x f_{X|A}(x) dx. \end{aligned}$$

Example 6.20 (The exponential random variable is memoryless). Alvin goes to a bus stop where the time T between two successive buses has an exponential PDF with parameter λ . Suppose that Alvin arrives t seconds after the preceding bus arrival and let us express this fact with the event $A = \{T > t\}$. Let X be the time that Alvin has to wait for the next bus to arrive. What is the conditional CDF $F_{X|A}(x)$?



$$\begin{aligned}
P[X > x|A] &= P[T > t + x|T > t] \\
&= \frac{P[T > t + x \text{ and } T > t]}{P[T > t]} \\
&= \frac{P[T > t + x]}{P[T > t]} \\
&= \frac{\exp^{-\lambda(t+x)}}{\exp^{-\lambda t}} \\
&= \exp^{-\lambda x}
\end{aligned}$$

Example 6.21. Suppose the duration T (in minutes) of a telephone call is an exponential $(1/3)$ random variable with the following PDF

$$f_T(t) = \begin{cases} (1/3)e^{-t/3}, & t \geq 0; \\ 0, & \text{otherwise.} \end{cases}$$

For calls that last at least 2 minutes, what is the conditional PDF of the call duration?

$$P[A] = P[T > 2] = \int_2^{\infty} f_T(t) dt = \int_2^{\infty} \frac{1}{3} e^{-t/3} dt = \frac{1}{3} \left(\frac{-3}{1} \right) e^{-t/3} \Big|_2^{\infty} = e^{-2/3}.$$

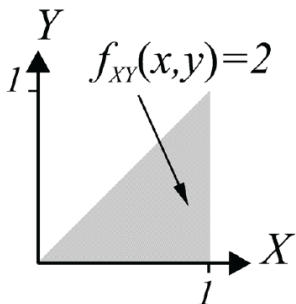
$$\begin{aligned} f_{T|T>2}(t) &= \begin{cases} \frac{f_T(t)}{P[T>2]}, & t \geq 2; \\ 0, & otherwise, \end{cases} \\ &= \begin{cases} \frac{1}{3} e^{-(t+2)/3}, & t \geq 2; \\ 0, & otherwise \end{cases} \end{aligned}$$

Conditioned on another Random Variable

Definition 6.12. If X and Y have a joint probability density function $f_{XY}(x, y)$, then the *conditional probability distribution* of X , given that $Y = y$, is defined for all values of y such that $f_Y(y) > 0$, by

$$f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)}.$$

Example: Suppose random variables X and Y are jointly uniformly distributed within the region $0 \leq y \leq x \leq 1$. Find the conditional PDFs $f_{X|Y}(x|y)$ and $f_{Y|X}(y|x)$.



$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)dy = \int_0^x 2dy = 2x$$

The conditional PDF of Y given X is

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} = \begin{cases} 1/x, & 0 \leq y \leq x; \\ 0, & \text{otherwise.} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)dx = \int_y^1 2dx = 2(1-y).$$

Therefore, the conditional PDF of X given Y is

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \begin{cases} 1/(1-y), & y \leq x \leq 1; \\ 0, & \text{otherwise.} \end{cases}$$

Product Rule or Law of Multiplication

$$\begin{aligned}f_{XY}(x, y) &= f_{X|Y}(x|y)f_Y(y) \\ &= f_{Y|X}(y|x)f_X(x)\end{aligned}$$



Example 6.23. First, a point Y is selected at random from the interval $(0, 1)$. Then another point X is chosen at random from the interval $(0, Y)$. Find the probability density function of X .

$$f_Y(y) = \begin{cases} 1, & 0 < y < 1; \\ 0, & \textit{otherwise}. \end{cases}$$

$$f_{X|Y}(x|y) = \begin{cases} 1/y, & 0 < y < 1, 0 < x < y; \\ 0, & \textit{otherwise}. \end{cases}$$

$$f_{XY}(x, y) = f_{X|Y}(x|y)f_Y(y) = \frac{1}{y} \cdot 1 = \frac{1}{y}$$

Sum Rule or Law of Total Probability

$$f_X(x) = \sum_i f_{X|A_i}(x)P[A_i]$$

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{XY}(x, y) dy \\ &= \int_{-\infty}^{\infty} f_{X|Y}(x|y) f_Y(y) dy \end{aligned}$$

Example 6.24. Assume that random variable X represents the voltage at the receiver in a modem. When symbol “0” is transmitted (event A_0), X is Gaussian $(-5, 2)$ random variable. When symbol “1” is transmitted (event A_1), X is the Gaussian $(5, 2)$ random variable. Given that symbols “0” and “1” are equally likely to be sent, what is the PDF of X ?

$$P[A_0] = P[A_1] = \frac{1}{2}.$$

$$f_{X|A_0}(x) = \frac{f_{XY}(x, y)}{P[A_0]} = \frac{1}{\sqrt{2\pi}} e^{-(x+5)^2/8} \cdot \frac{1}{2} = \frac{1}{2\sqrt{2\pi}} e^{-(x+5)^2/8}$$

$$f_{X|A_1}(x) = \frac{f_{XY}(x, y)}{P[A_1]} = \frac{1}{2\sqrt{2\pi}} e^{-(x-5)^2/8}.$$

$$\begin{aligned} f_X(x) &= f_{X|A_0}(x)P[A_0] + f_{X|A_1}(x)P[A_1] \\ &= \frac{1}{4\sqrt{2\pi}} \left(e^{-(x+5)^2/8} + e^{-(x-5)^2/8} \right) \end{aligned}$$

Example 6.25. The PDF of random variable X and the conditional PDF of random variable Y given X are

$$f_X(x) = \begin{cases} 3x^2 & 0 \leq x \leq 1, \\ 0 & \text{otherwise,} \end{cases}$$

and

$$f_{Y|X}(y|x) = \begin{cases} 2y/x^2 & 0 \leq y \leq x, 0 < x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Find the joint PDF of X and Y , and the marginal PDF of Y .

$$f_{XY}(x, y) = f_{Y|X}(y|x)f_X(x) = \begin{cases} 6y & 0 \leq y \leq x, 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} f_Y(y) &= \int_y^1 f_{Y|X}(x|y)f_X(x)dx \\ &= \int_y^1 \frac{2y}{x^2} \cdot 3x^2 dx \\ &= 6xy \Big|_y^1 \\ &= 6y(1 - y) \end{aligned}$$

Conditional Expectation

$$E[X|A] = \int_{-\infty}^{\infty} x f_{X|A}(x) dx$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_{X|A}(x) dx$$

Example 6.26. Find the conditional expectation $E[T|T > t]$ for the problem in Example 6.21.

$$f_T(t) = \begin{cases} (1/3)e^{-t/3}, & t \geq 0; \\ 0, & \textit{otherwise}. \end{cases}$$

$$\begin{aligned} f_{T|T>2}(t) &= \begin{cases} \frac{f_T(t)}{P[T>2]}, & t \geq 2; \\ 0, & \textit{otherwise}, \end{cases} \\ &= \begin{cases} \frac{1}{3}e^{-(t+2)/3}, & t \geq 2; \\ 0, & \textit{otherwise}. \end{cases} \end{aligned}$$

$$E[T|T > t] = \int_2^\infty t \frac{1}{3} e^{-(t-2)/3} dt$$

Integration by parts yields

$$E[T|T > 2] = -te^{-(t-2)/3} \Big|_2^\infty + \int_2^\infty e^{-(t-2)/3} dt = 2 + 3 = 5$$

$$E[X|Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx.$$

$$E[g(X, Y)|Y = y] = \int_{-\infty}^{\infty} g(x, y) f_{X|Y}(x|y) dx$$

Example 6.27. For random variables X and Y the conditional PDF is given by

$$f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)} = \begin{cases} 1/(1-y) & y \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Find the conditional expected values $E[X|Y = y]$ and $E[X|Y]$.

$$\begin{aligned} E[X|Y = y] &= \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx \\ &= \int_y^1 \frac{1}{1-y} x dx \\ &= \left. \frac{x^2}{2(1-y)} \right|_{x=y}^{x=1} \\ &= \frac{1+y}{2}. \end{aligned}$$

Independence

Two events A and B are independent if

$$P[AB] = P[A]P[B]$$

Two RVs X and Y are independent if, for any subsets A and B of real numbers, the events $\{X \in A\}$ and $\{Y \in B\}$ are independent,

$$P[X \in A, Y \in B] = P[X \in A]P[Y \in B]$$

Which can be easily extended, for any two real numbers x and y , to

$$P[X \leq x, Y \leq y] = P[X \leq x]P[Y \leq y]$$

If X and Y are two RVs defined on the same sample space, then X and Y are independent iif, for all real numbers x and y ,

$$F_{XY}(x, y) = F_X(x)F_Y(y)$$

And is applicable for both discrete and continuous RVs

Independence of Discrete RVs

If X and Y are two discrete RVs with set of possible values S_X and S_Y , respectively, then they are independent if,

- $\forall x \in S_X$ and $\forall y \in S_Y$,
- the events $\{X = x\}$ and $\{Y = y\}$ are independent
- which, in turn, satisfies

$$P[X = x, Y = y] = P[X = x]P[Y = y]$$

$$P_{XY}(x, y) = P_X(x)P_Y(y)$$

In contrast, if X and Y are two discrete independent RVs with set of possible values S_X and S_Y , then

$$P[X = x|Y = y] = P[X = x]$$

i.e., knowing that $Y = y$, does not change the Probability of $\{X = x\}$

Example: Suppose there are 10 red and 30 blue balls in an urn. Suppose further that you first pick 13 balls from the urn without replacement, and then pick 20 balls without replacement. Find the probability that you pick 4 red balls among the first 13 balls and 7 red balls among the next 20 balls.

$X \triangleq$ # of red balls among the first 13 balls

$Y \triangleq$ # of red balls among the second 20 balls

$X \sim$

$Y \sim$

$P[X = 4, Y = 7] =$

Can be found from $P_{XY}(x, y)$

$$P_{XY}(4,7) = \binom{13}{4} \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^9 \binom{20}{7} \left(\frac{1}{4}\right)^7 \left(\frac{3}{4}\right)^{13}$$

$$\begin{aligned} P_{XY}(x,y) &= \binom{n}{x} (p)^x (1-p)^{n-x} \binom{m}{y} (p)^y (1-p)^{m-y} \\ &= \binom{n}{x} \binom{m}{y} p^{x+y} (1-p)^{(n+m)-(x+y)} \end{aligned}$$

If X and Y are two independent RVs, then for real valued functions

$$g: R \rightarrow R \text{ and } h: R \rightarrow R$$

$g(X)$ and $h(Y)$ are also independent RVs.

If X and Y are two independent RVs, then for real valued functions

$$g: R \rightarrow R \text{ and } h: R \rightarrow R$$

$$E[g(X)h(Y)] = E[g(X)]Eh(Y)]$$

A special case of which

$$E[XY] = E[X]E[Y]$$

However, the converse is not true. Above equation does not guarantee the independence of X and Y

Independence of Continuous RVs

If X and Y are jointly continuous RVs with the JPDF $f_{XY}(x, y)$, then they are independent, iif

- $\forall x \in S_X$ and $\forall y \in S_Y$
- The JPDF is equal to the product of the marginal PDFs, $f_X(x)$ and $f_Y(y)$
- $f_{XY}(x, y) = f_X(x)f_Y(y)$

Example: Suppose there are two paths, A and B , between two computer nodes in a network. The PDF of the delays of packets, in hundreds of microseconds, for the two paths, is given by

$$f(x) = \begin{cases} \frac{x}{4}, & 1 < x < 3 \\ 0, & \text{otherwise.} \end{cases}$$

If the packet delay of one path is independent of the other, find the probability that, for two packets sent in two paths, the delay of one path is at least 50 microseconds more than the other path.

$X \triangleq$ delay of path A

$Y \triangleq$ delay of path B

The desired probability is

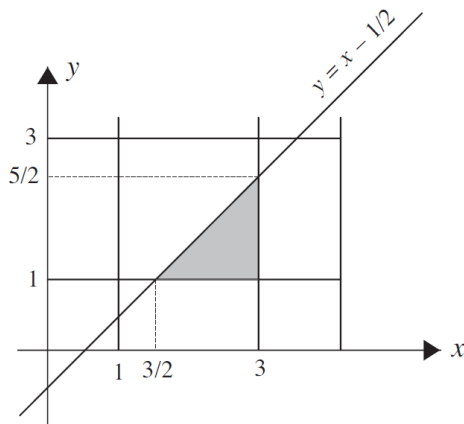
$$P\left[X > Y + \frac{1}{2}\right] + P\left[Y > X + \frac{1}{2}\right]$$

$$f_{XY}(x, y) = f_X(x)f_Y(y)$$

$$f_X(x) =$$

$$f_Y(y) =$$

$$f_{XY}(x, y) =$$



$$\begin{aligned}
 2P\left(X > Y + \frac{1}{2}\right) &= 2P\left((X, Y) \in \left\{(x, y): \frac{3}{2} < x < 3, 1 < y < x - \frac{1}{2}\right\}\right) \\
 &= 2 \int_{3/2}^3 \left(\int_1^{x-1/2} \frac{xy}{16} dy \right) dx = \frac{1}{8} \int_{3/2}^3 \left[\frac{xy^2}{2} \right]_1^{x-1/2} dx \\
 &= \frac{1}{16} \int_{3/2}^3 x \left[\left(x - \frac{1}{2}\right)^2 - 1 \right] dx \\
 &= \frac{1}{16} \int_{3/2}^3 \left(x^3 - x^2 - \frac{3}{4}x \right) dx \\
 &= \frac{1}{16} \left[\frac{1}{4}x^4 - \frac{1}{3}x^3 - \frac{3}{8}x^2 \right]_{3/2}^3 = \frac{549}{1024} \approx 0.54. \quad \blacklozenge
 \end{aligned}$$